

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Fifth Semester B.E. (Biotechnology) (Credit System) Degree Examinations**

January 2023

20BTC113 - FOOD BIOTECHNOLOGY

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing Two full questions from Unit – I & Unit – II each and One full question from Unit – III.

		Marks	BT*	CO*	PO*
Unit – I					
1.	a) Describe how food quality will affect the health.	10	L*1	1	1
	b) Explain in details about primary sources of microbes in food.	10	L2	2	2
2.	a) Give detail explanation on microbial spoilage of fruits and vegetables.	10	L1	1	1
	b) Illustrate the effect of freezing and thermal destruction on microorganisms with its working principle.	10	L1	2	2
3.	a) What are chemical food preservatives? Add a note on Benzoic acid and sulphur dioxide.	10	L1	1	1
	b) Discuss the different types of non bacterial food borne diseases.	10	L2	2	2
Unit – II					
4.	a) Explain briefly the conventional detection methods of microbes and their products in food.	10	L1	3	1
	b) Illustrate how <i>Clostridium parfringens</i> and <i>vibrio</i> species are detected and confirmed in food.	10	L2	4	2
5.	a) Write a short note on i) Prebiotics and ii) Probiotics.	10	L1	3	1
	b) Elucidate the role of different enzymes in food industries.	10	L1	4	2
6.	a) Give details on effect of food processing on food constituents.	10	L1	3	1
	b) What is meant by Genetically modified food? Explain briefly Golden rice and BT Brinjal.	10	L2	4	2
Unit – III					
7.	a) Explain the industrial production of citric acid.	10	L1	5	1
	b) What are single cell proteins? Explain the occurrence, processing and importance of spirulina.	10	L2	5	1
8.	a) Describe the detail production of cheese.	10	L1	5	2
	b) Write a short note on Kimachi and Sauerkraut.	10	L2	5	2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
